Unit 3 Progress Check: FRQ Part B

1. The Car class, which you will write, is used to represent information about a car.

Car objects are created by calls to a constructor with two double parameters. The first parameter represents the number of gallons of fuel in the car. The second parameter represents the average number of miles the car can travel per gallon of fuel.

The Car class contains a canMakeTrip method, which determines whether there is enough fuel in the car to make a trip. This method has a double parameter that represents the distance, in miles, of a potential trip. If there is enough fuel, the method updates the current amount of fuel in the car and returns true. If there is not enough fuel for the entire distance of the trip, the amount of fuel is unchanged and the method returns false.

The following table contains a sample code execution sequence and the corresponding results. The code execution sequence appears in a class other than Car.

Statement	Return Value (blank if no value)	Explanation
boolean possible;		
Car carOne = new Car(10.0, 30.0);		A Car object named CarOne is constructed with 10 gallons of fuel and an average of 30 miles per gallon.
possible = carOne.canMakeTrip(120.0);	true	At 30 miles per gallon, a 120-mile trip will require 4 gallons of fuel. Since this is less than or equal to the 10 gallons of fuel remaining, the trip can be made. The amount of fuel remaining will be 6 gallons.
possible = carOne.canMakeTrip(90.0);	true	At 30 miles per gallon, a 90-mile trip will require 3 gallons of fuel. Since this is less than or equal to the 6 gallons of fuel remaining, the trip can be made. The amount of fuel remaining will be 3 gallons.
possible = carOne.canMakeTrip(225.0);	false	At 30 miles per gallon, a 225-mile trip will require 7.5 gallons of fuel. Since this is more than the 3 gallons of fuel remaining, the trip cannot be made and the amount of fuel is unchanged.
Car carTwo = new Car(12.0, 15.0);		A Car object named carTwo is constructed with 12 gallons of fuel and an average of 15 miles per gallon.
possible = carTwo.canMakeTrip(300.0);	false	At 15 miles per gallon, a 300-mile trip will require 20 gallons of fuel. Since this is more than the 12 gallons of fuel remaining, the trip cannot be made and the amount of fuel is unchanged.
possible = carTwo.canMakeTrip(180.0);	true	At 15 miles per gallon, a 180-mile trip will require 12 gallons of fuel. Since this is less than or equal to the 12 gallons of fuel remaining, the trip can be made. The amount of fuel remaining will be 0 gallons.

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Write the complete `Car` class. Your implementation must meet all specifications and conform to the examples shown in the table.

2. The TrashCan class, which you will write, represents a trash can.

TrashCan objects are created by calls to a constructor with a double parameter that represents the trash can's capacity, which is the maximum weight, in pounds, that a TrashCan object can hold. Assume that this value will be greater than or equal to 0. When a `Trashcan` object is constructed, it is initially empty and its contents have a weight of 0 pounds.

The TrashCan class contains an acceptTrash method, which has a double parameter that represents the weight, in pounds, of the trash that will be deposited into the trash can. The amount to be deposited will always be greater than 0 and less than the capacity of the can. If there is already trash in the trash can, the amount to be deposited, combined with the existing trash, may reach or exceed the can's capacity. In this case, the trash can will be filled to its capacity and then emptied before the remaining trash is accepted. The trash can should always be emptied when the weight of the trash in the can reaches the can's capacity. The acceptTrash method returns a double that represents the number of pounds of trash that can still be added to the trash can before it is completely filled.

The following table contains a sample code execution sequence and the corresponding results. The code execution sequence appears in a class other than TrashCan.

Statement	Return Value (blank if no value)	Explanation
double remaining;		
TrashCan kitchen = new TrashCan(10.0);		A TrashCan object named kitchen is constructed with a capacity of 10 pounds of trash. Initially there are 0 pounds of trash in the trash can.
remaining = kitchen.acceptTrash(2.5);	7.5	kitchen now holds 2.5 pounds of trash and can take 7.5 more pounds before it needs to be emptied.
remaining = kitchen.acceptTrash(3.5);	4.0	kitchen now holds 6 pounds of trash and can take 4 more pounds before it needs to be emptied.
remaining = kitchen.acceptTrash(6.0);	8.0	kitchen accepts 4 pounds of trash, reaching the 10-pound capacity, and is then emptied. The remaining 2 pounds of trash are then accepted. kitchen now holds 2 pounds of trash and can take 8 more pounds before it needs to be emptied.
TrashCan bedroom = new TrashCan(3.0);		A TrashCan object named bedroom is constructed with a capacity of 3 pounds of trash. Initially there are 0 pounds of trash in the trash can.

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Statement	Return Value (blank if no value)	Explanation
remaining = bedroom.acceptTrash(1.0);	2.0	bedroom now holds 1 pound of trash and can take 2 more pounds before it needs to be emptied.
remaining = bedroom.acceptTrash(3.2);	1.8	bedroom accepts 2 pounds of trash to reach the 3-pound capacity and then is emptied. The remaining 1.2 pounds of trash are then accepted. bedroom now holds 1.2 pounds of trash and can take 1.8 more pounds before it needs to be emptied.

Write the complete TrashCan class. Your implementation must meet all specifications and conform to the examples shown in the table.